DEEP-DIVE ARCHITECTURAL & LOGIC AUDIT #04D: WEAPONS ARSENAL (PART 4 — GRAND DAO TACTICAL & ESOTERIC IMPLEMENTS: FAN, ASTROLABE, ABACUS)

- **Target Codebase**: `JianghuSect 2` (`JS2-v2.6.0` / `SIM.PATCH_VERSION = '2.6.0'` / `HC.FINGERPRINT = '155:2f9c9dd7'`)
- **Target Audience**: Autonomous AI Software Engineers (**GPT** & **Gemini** Models)
- **Weapon Audit Series Plan (`04A–04E`)**:
  - `04A`: Group 1 — Core Bladed & Polearm Melee (`SWORD`, `BLADE`, `SPEAR`, `STAFF`)
  - `04B`: Group 2 — Brawling, Ballistic & Flexible (`FIST`, `CLAW`, `BOW`, `NEEDLE`, `WHIP`)
  - `04C`: Group 3 — Scholarly, Musical & Dharma Conduits (`FOCUS`, `ZITHER`, `BRUSH`, `ROSARY`)
  - **`04D` (This Document)**: Group 4 — **Grand Dao Tactical & Esoteric Implements (`FAN` พัดยุทธ์/พัดเทวพายุ, `ASTROLABE` จานดารา/หลั่วผานธรณี, `ABACUS` ลูกคิดทองคำพิพากษา)**
  - `04E`: Part 5 — Holistic Weapon Ecosystem & Cross-System Catch-All Audit

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0. Hard Architectural Invariants for AI Implementers

1. **Immutable `human_core.js`**: Never modify `GAME_SOURCE/js/human_core.js` (`HC.FINGERPRINT === '155:2f9c9dd7'`). Proficiency keys `FAN`, `ASTROLABE`, and `ABACUS` must be read/written strictly via `M.skillRead` / `M.skillAdd` / `M.skillPractice`.
2. **Every Applied Status Must Have a Real Consumer**: Never push decorative dummy strings (`'tempoBreak'`, `'contractDefault'`, `'gravityCrush'`) onto `unit.sx` without wiring them into `STATS.get`, `STATS.explain`, `COMBAT`, `M.combatTargetBonus`, and `EFX_CLEAR_DEBUFFS`!
3. **100% Emergent NPC Autonomy**: Strategists, Weather Callers, Diviners, Geomancers, Treasurers, and Merchant Magnates must naturally value, craft, discover, and wield their iconic implements in both combat and daily sect governance.

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1. [WPN-STATUS-01] Critical Discovery: tempoBreak, contractDefault & gravityCrush on Legendary FAN, ABACUS & ASTROLABE Are 100% Inert Dummy Strings!

Full-repository verification across `js/visitors_clues.js`, `js/skills.js`, `js/stats.js`, `js/combat.js`, and `js/martial.js` proves a **major silent failure**:

1. **`tempoBreak` (**สลายจังหวะ **— `LEG_WPN_BAJIAO_TEMPEST_FAN`, `visitors_clues.js:417, 882`)**:
   - `VC.runWeaponProcs` calls `ST.addStatus(defender, 'tempoBreak', 7)`.
   - `SK.STATUS.tempoBreak` (`skills.js`) defines its display label/icon, **but NEITHER `stats.js` NOR `combat.js` contains a single reference to `'tempoBreak'`**! It modifies **`0` stats** and delays **`0` cooldowns**!
2. **`contractDefault` (**ผิดนัดพันธสัญญา **— `LEG_WPN_CAISHEN_GOLDEN_ABACUS`, `visitors_clues.js:469, 882`)**:
   - `VC.runWeaponProcs` calls `ST.addStatus(defender, 'contractDefault', 8)`.
   - **NEITHER `stats.js` NOR `combat.js` contains a single reference to `'contractDefault'`**! It modifies **`0` stats** and triggers **`0` foreclosure effects**!
3. **`gravityCrush` (**แรงโน้มถ่วงกดทับ **— `LEG_WPN_HETU_LUOSHU_ASTROLABE`, `visitors_clues.js:443, 882`)**:
   - In `v2.4.0` (`DA-011`), `applyGravityField` was patched to apply `'gravity_slow'`, **leaving `HETU_LUOSHU_ASTROLABE` still pushing `{ k: 'gravityCrush' }` into `defender.sx`**, which `stats.js` never reads (while `combat.js` checks property `u.gravityCrush`, NOT `ST.has(u, 'gravityCrush')`)!
4. **Excluded From `M.combatTargetBonus` (`martial.js:390`) & `EFX_CLEAR_DEBUFFS` (`economy.js:918`)**:
   - All 7 `v2.0.0` debuffs (`meridianSeal`, `tempoBreak`, `guLeech`, `gravityCrush`, `contractDefault`, `voidRiftTear`, `entropyDecay`) are missing from `martial.js:390`'s `targetDebuffed` whitelist (breaking target-debuff passives!) and from `economy.js:918`'s cleanse list!

Exact Unified Status Wiring Blueprint:

- **`tempoBreak`**: In `stats.js` (`compute` & `getBaseFast`), apply `cast -= 24`, `acc -= 12`; on initial application in `combat.js` / `runWeaponProcs`, push active skill cooldowns back by `+2` ticks!

- **`contractDefault`**: In `stats.js`, apply `def *= 0.82`, `mdef *= 0.82`, `eva -= 15`; in `combat.js`, `MERCANTILE_DAO` / `ABACUS` hits against `contractDefault` deal `+16%` foreclosure damage and generate `+10 CONTRACT_LEDGER`!
- **`gravityCrush`**: Alias/unify `ST.has(u, 'gravityCrush') || ST.has(u, 'gravity_slow') || u.gravityCrush > 0` in **both** `stats.js` (`move *= 0.50`, `eva *= 0.70`) and `COMBAT.moveToward` (grounding flight & blocking `blink`/`dash` on both `foe` and `!foe` units, respecting `bossImmune`)!
- **`meridianSeal`** / **`voidRiftTear`** / **`entropyDecay`** / **`guLeech`**: Wire concrete stat reductions in `stats.js` (`meridianSeal`: `-22 cast` & blocks passive Qi regen; `voidRiftTear`: `-20% def/mdef`; `entropyDecay`: `-15%` all elemental resistances; `guLeech`: periodic Qi-to-HP leech in `statusTick`) and add all 7 to `targetDebuffed` (`martial.js:390`) and `EFX_CLEAR_DEBUFFS` (`economy.js:918`)!

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2. `[WPN-FAN-02]` Weapon Family #14: `FAN` (พัดเทวพายุ / พัดขนนกกุนซือ / พัดโครงเหล็กซ่อนคม)

2.1 Untranslated Hanzi in Thai Name, Zero Clue Quest, & Missing Zhuge-Style Strategists!

- **Locations**: `js/visitors_clues.js:396–422`, `js/martial.js`
- **Four Concrete Bugs**:
  1. **Raw Chinese Characters Inside `nameTh` (`visitors_clues.js:397`)**:
     - `nameTh: 'พัดเทวะ芭蕉โบกสะบัดสี่ฤดู'` leaves raw Hanzi 芭蕉 untranslated inside the Thai display string! Rename `nameTh` to `'พัดเทวะใบกล้วยทิพย์โบกสะบัดสี่ฤดู'` (preserving `chineseName: '太虛芭蕉四时扇'`).
  2. **Zero Clue Template in `VC.CLUE_TEMPLATES`**:
     - Despite `obtainSources: ['CLUE_QUEST', ...]`, `LEG_WPN_BAJIAO_TEMPEST_FAN` has **zero** clue quest! Author `CLUE_LEG_WPN_BAJIAO_TEMPEST_FAN` (ยอดเขาเพลิงพายุและพัดเทวะใบกล้วยทิพย์สี่ฤดู — surviving seasonal gale trial + `FAN` / `intellect.comprehension >= 60`).
  3. **Strategists (`Zhuge Feather Fan`!) Omitted From `FAN` Profiles & `BAJIAO_TEMPEST_FAN`**:
     - `visitors_clues.js:400` explicitly cites *"Zhuge Feather Fan Lore (羽扇)"*, **yet `STRATEGY_GOVERNANCE` (`BATTLEFIELD_STRATEGIST`, `SHADOW_INTELLIGENCE_MARSHAL`) and `SAGE_SUPPORT.EIGHT_TRIGRAMS_BANNER_SAGE` do NOT include `'FAN'` in their preferred weapons or `BAJIAO_TEMPEST_FAN.compatiblePath` / `preferredSubs`!**

- Meanwhile, `BAJIAO_TEMPEST_FAN.preferredSubs` (`line 411`) mistakenly lists `'VERMILION_PHOENIX_WING_ASCENDANT'` (`PRIMORDIAL_SHIFTER` — a `CLAW` shifter whose path isn't even in `compatiblePath`!) and suffers from the `'THUNDER_CLOUD_CALLER'` vs `'THUNDERSTORM_CALLER'` alias split!

4. **Missing Gale Knockback & Iron-Ribbed Fan Subtype**:
   - Wire `EFX_V200_TEMPEST_FAN_GALE` into `COMBAT runItemEffectProcs` so `FAN` crits/procs push encroaching melee enemies back `+1.2` tiles (kiting self-peel!) and boost `CELESTIAL_WEATHER SKY_QI` gain by `+15%`.

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3. `[WPN-ASTROLABE-03]` Weapon Family #15: ASTROLABE (จานสวรรค์ดารา / จานหลัวผานธรณี)

3.1 Legacy Star/Array/Oracle Subpaths Blind to ASTROLABE (`js/martial.js:31, 46, 65`)

- Pre-`v2.0.0` astrological and formation subpaths were never updated to recognize `ASTROLABE`:
  - `XIAN.ASTRAL_ARRAY` (สายค่ายกลดารา): `['FOCUS', 'SWORD']` $\to$ add `'ASTROLABE'`!
  - `GOD.FATE_ORACLE` (สายเทพชะตาพยากรณ์): `['FOCUS', 'STAFF', 'BOW']` $\to$ add `'ASTROLABE'`!
  - `MECH_PUPPET.ASTRAL_CLOCKWORK_PRIME` (สายสมองกลดาราพยากรณ์): `['FOCUS', 'SWORD']` $\to$ add `'ASTROLABE'`!
  - `ARRAY_TALISMAN.EARTH_VEIN_WARDEN` & `ASTRAL_STARFALL_ARRAY` $\to$ add `'ASTROLABE'`!
- Also add these subpaths to `LEG_WPN_HETU_LUOSHU_ASTROLABE.preferredSubs` and `M.SUB_EQUIP_BONUSES`, and author `CLUE_LEG_WPN_HETU_LUOSHU_ASTROLABE` (หอดูดาวร้างกลางแม่น้ำลั่วและจานสวรรค์เหอถูลั่วซู — `ARRAY`/`ASTROLABE` + `intellect.reasoning >= 62` trial)!

3.2 Slot-Guard Collision With ACCESSORY Array Disks (จานค่ายกล / เข็มทิศดารา) & World Utility

- In `data_items.js`, `ITEM190_CORE_001` (จานค่ายกลหยกประสานห้าธาตุ), `ITEM190_CORE_007` (จานแม่เหล็กสะกดกระบี่บิน), `REC140_ARR_MIST_DISK_2` (จานค่ายกลหมอกบังตา), and `ITEM190_TOOL_003` (เข็มทิศดารา) all have `slot: 'ACCESSORY'`, while `ASTROLABE` (`จานหลัวผาน...` / `จานคำนวณดารา...`) has `slot: 'WEAPON'`.

- Ensure `JD.MARTIAL.classify(slot, nameTh)` checks `slot` strictly before matching `/จาน|หลัวผาน|เข็มทิศ/`, and grant equipped `ASTROLABE` weapons `+12%` `explore.js` hazard/trap avoidance + `+10%` `FENG_SHUI` / `LEYLINE_GRID` anchoring efficiency + `EFX_V200_ASTROLABE_STAR_LOCK` on-hit proc in `combat.js`!

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4. [WPN-ABACUS-04] Weapon Family #16: ABACUS (ลูกคิดทองคำพิพากษา / ลูกคิดสำริดตราชั่ง)

4.1 Zero Clue Quest, Missing Steward/Logistics/Audit Subpaths, & Zero Economic Governance Bonus!

1. **Zero Clue Template in `VC.CLUE_TEMPLATES`**:
   - Author `CLUE_LEG_WPN_CAISHEN_GOLDEN_ABACUS` (คลังสมบัติสมาคมวานิชโบราณและมหาลูกคิดทองคำไฉ่ซิง — `trading` appraisal + `ABACUS` / `personality.integrity >= 60` ledger trial).
2. **Missing Steward, Logistics & Karmic Auditor Subpaths (`js/martial.js`)**:
   - Add `'ABACUS'` to the preferred weapon lists (`M.PROFILES` & `M.SUB_EQUIP_BONUSES`) and `CAISHEN_GOLDEN_ABACUS.preferredSubs` for:
     - `STRATEGY_GOVERNANCE`: `SECT_STEWARD` (ผู้ดูแลรากฐานสำนัก), `CARAVAN_QUARTERMASTER` (ผู้จัดเสบียงคาราวาน), `GRAND_LOGISTICS_COMMANDER` (แม่ทัพพลาธิการ).
     - `KARMA_SAMSARA`: `CAUSE_EFFECT_LEDGER` (ผู้ถือสมุดบัญชีเหตุผลแห่งกรรม).
3. **Zero Out-of-Combat Economic & Anti-Corruption Utility**:
   - An Abacus is both a hidden-projectile/acupoint weapon (ซัดเม็ดลูกคิดทองคำทะลวงเกราะ) and a master accountant's calculating engine!
   - Wire equipped `ABACUS` (`tier >= 3`) into:
     - `external_market.js` (`EXT.tradeSect` / `EXT.tradePersonal`): `+1.5% * tier` better trade spread when worn by the acting trader/`merchant` officer.
     - `sect.js` (`SECT.audit` by `censor` / `treasury` / `master`): `+15%` embezzlement detection accuracy (คำนวณบัญชีคลังไม่คลาดเคลื่อนแม้แต่ตำลึงเดียว)!
     - `combat.js` (`runItemEffectProcs`): Wire `EFX_V200_ABACUS_LEDGER_SETTLE` so on-hit procs apply `contractDefault` and cash out bonus damage proportional to debuff stacks on the target!

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5. [WPN-NEW6-PARITY-05] Officer & Aspiration Blindness Toward All 6 v2.0.0 Weapons (sect.js & cognition.js)

- **js/sect.js (observableCapability line 76 & SECT.relevance line 260)** and **js/cognition.js (evalAspiration 'martial_master' line 319)** still compute weapon mastery from the hardcoded 7 legacy keys (SWORD, BLADE, SPEAR, FIST_PALM, BOW, STAFF, FOCUS)—giving **0 martial credit** to Grandmasters of FAN, ASTROLABE, ABACUS, ZITHER, BRUSH, WHIP, ROSARY, CLAW, and NEEDLE!
- Centralize `M.ALL_WEAPON_SKILL_KEYS = ['SWORD','BLADE','SPEAR','FIST_PALM','BOW','STAFF','FOCUS','ZITHER','BRUSH','WHIP','ROSARY','CLAW','NEEDLE','FAN','ASTROLABE','ABACUS']` and `M.maxWeaponSkill(n)` so every system in the game treats all 16 weapon families as first-class citizens!

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6. Implementation & Verification Checklist for GPT / Gemini (04D)

- [ ] **Wire All 7 v2.0.0 Debuffs (stats.js, combat.js, martial.js, economy.js)**: Make tempoBreak, contractDefault, gravityCrush, meridianSeal, voidRiftTear, entropyDecay, and guLeech apply real stat/cooldown/movement effects, trigger M.combatTargetBonus, and be cleansable by EFX_CLEAR_DEBUFFS.
- [ ] **FAN**: Fix raw Hanzi in LEG_WPN_BAJIAO_TEMPEST_FAN.nameTh; author CLUE_LEG_WPN_BAJIAO_TEMPEST_FAN; add Zhuge-style STRATEGY_GOVERNANCE strategists & fix THUNDERSTORM_CALLER alias; wire EFX_V200_TEMPEST_FAN_GALE knockback in combat.
- [ ] **ASTROLABE**: Unify gravityCrush on HETU_LUOSHU_ASTROLABE; author CLUE_LEG_WPN_HETU_LUOSHU_ASTROLABE; add 'ASTROLABE' to ASTRAL_ARRAY, FATE_ORACLE, ASTRAL_CLOCKWORK_PRIME, EARTH_VEIN_WARDEN, ASTRAL_STARFALL_ARRAY; add Luopan exploration/Feng-Shui bonus.
- [ ] **ABACUS**: Author CLUE_LEG_WPN_CAISHEN_GOLDEN_ABACUS; add 'ABACUS' to SECT_STEWARD, CARAVAN_QUARTERMASTER, GRAND_LOGISTICS_COMMANDER, CAUSE_EFFECT_LEDGER; wire EXT trade margin, SECT.audit anti-corruption bonus, and EFX_V200_ABACUS_LEDGER_SETTLE combat proc; unify M.ALL_WEAPON_SKILL_KEYS across sect.js and cognition.js.